Latent Coherence: Do Frontier Language Models' Probability Judgments Obey the Law of Total Probability?

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October 2026

Abstract

A model that reasons by weighing hidden explanations of its evidence, as LawZero's *Scientist AI* proposal envisions, needs probability judgments that fit together. We test whether frontier language models' judgments satisfy the law of total probability, $P(X) = \sum_z P(z)\,P(X \mid z)$, when the latent variable z must be inferred from records in the context. Each item gives a model a short set of records (source track records, an inspection log, a count table, or a fictionalized multi-hop claim from HoVer) and asks, in separate single-turn calls, for $P(X)$, for $P(z)$ over the latent's states, and for $P(X \mid z)$. One synthetic family, *causal*, applies the same test to $P(y \mid do(x))$ and the back-door adjustment. Every query has an exact ground-truth probability, so coherence and accuracy are measured separately, and a subset of items is asked in three wordings so that incoherence that *persists* across wordings can be separated from rewording noise. Across seven models and 1,590 shared prompts per model, GPT-6.1 Sol and Claude Opus 5.5 are the most coherent (mean absolute residual 0.08 and 0.12 log-odds). Claude Haiku 5.5, Gemini 3.8 Flash and DeepSeek V4-Pro show substantial persistent incoherence (0.27–0.35 log-odds, with 95% intervals excluding zero). Their incoherence is mostly item-specific spread rather than a shared offset, and the small leans that remain point in opposite directions. Within every model, instances with larger residuals have larger errors against the truth (Spearman 0.40–0.62), so the residual, which needs no ground truth, flags unreliable answers. Reasoning length does not predict coherence, either across models (the most coherent used 400–500 reasoning tokens per answer, three less coherent ones 4,400–7,200) or within a model across rewordings of the same question. Simple consistency checks (normalization, the back-door identity) pass almost everywhere. The failures are specific to reasoning through a latent variable.

1 Introduction

Many proposals for safer AI systems lean on calibrated probabilistic judgment rather than goal-directed action. LawZero's *Scientist AI* [1] is a prominent example: a non-agentic system that explains data by positing latent causes and answers queries with probabilities obtained by averaging over those explanations. Averaging over latent explanations is the law of total probability,

$$P(X) \;=\; \sum_z P(z)\,P(X \mid z), \tag{1}$$

and a predictor that violates it gives answers that depend on how a question is decomposed. If such a system is built from, or amortized by, a language model, the model's probability judgments need to obey identities like (1), at least approximately.

We ask how close current frontier models come. Prior work has shown that language model probability judgments violate simple identities such as complementation and the conjunction rule [12], and that forecasters can be scored by the consistency of logically related forecasts [3, 8]. We target a specific and safety-relevant

case: coherence under marginalization over a latent variable that the model must *infer from evidence in context*. The latent is never stated. It might be whether a news source is careful, judged from its track record, or which of three faults is present, judged from an inspection log. Reasoning through such a hidden quantity is exactly what Scientist AI asks of its inference machine, and it is where we expect incoherence to concentrate.

Contributions.

- An eval of latent coherence with four item families (§3.2). Three are synthetic, with exact ground truth. The fourth is built from HoVer multi-hop claims [4], fictionalized so that models cannot answer from memory. One synthetic family, `causal`, tests $P(y \mid do(x))$ against the back-door adjustment.
- A noise-corrected decomposition of each model's residual into rewording noise and a *persistent* component (§3.3), so that unstable answers are not mistaken for incoherent judgments.
- Results for seven frontier models (§5). Two models are coherent to within about 2 percentage points on average. Three have persistent incoherence that varies from item to item. Large residuals mark inaccurate answers within every model, and which estimate breaks (the direct answer or the decomposition) differs by model. Reasoning length is not associated with coherence across models, nor within a model across rewordings of the same question.

The code (MIT), data (CC BY-SA 4.0) and evaluation harness are at github.com/ppotash/latent-coherence.

2 Related work

Coherence of probability judgments. Human probability judgments are famously incoherent, as in the conjunction fallacy [11]; Zhu et al. [13] explain such deviations as the product of sampling-based inference. Zhu and Griffiths [12] find the same signatures in LLMs, testing identities over events stated directly in the prompt. We test a different identity, marginalization over a latent the model must infer, and we separate wording noise from persistent incoherence instead of pooling them.

Consistency checks. Fluri et al. [3] evaluate models on tasks without ground truth by checking logical consistency, and Paleka et al. [8] turn this into arbitrage-based metrics for LLM forecasters. Our items have exact ground truth, which lets us relate incoherence to accuracy directly.

Probabilistic and causal reasoning benchmarks. QUITE [10] and BLInD [7] test Bayesian-network reasoning from text against computed answers. CLadder [5] tests causal queries across Pearl's ladder [9]. These benchmarks score accuracy. We score whether a model's *own* answers agree with each other, which needs no ground truth, and we report accuracy separately. Calibration work [6] asks whether stated confidence matches frequency, a property that is distinct from, and does not imply, coherence across related queries.

3 The eval

3.1 Identities

For an item with target event X and latent Z with states $z_1, \ldots, z_K$, a model is asked for $\hat{P}(X)$, $\hat{P}(Z = z_k)$ and $\hat{P}(X \mid Z = z_k)$ in $1 + 2K$ separate single-turn calls, so no answer is ever visible to another query. Its *decomposition* is $D = \sum_k \hat{P}(z_k)\,\hat{P}(X \mid z_k)$. The **residual** is the gap in log-odds,

$$r \;=\; \operatorname{logit}\hat{P}(X) - \operatorname{logit} D, \tag{2}$$

with probabilities clipped to $[0.01,\ 0.99]$. Between 6.6% and 9.6% of each model's answers lie at or beyond these bounds; the conclusions do not depend on the clip level or on the log-odds scale (Appendix D.2).

Family	Records in context	Latent Z
`source`	Reports on a claim from 1–3 sources, each with a track record of checked past reports	Whether one source is careful or careless
`diagnostic`	A fault-diagnosis inspection log, with symptom rates for each of three causes	Which of three causes is present
`causal`	A count table over confounder Z, treatment X and outcome Y	The confounder (e.g. which production line a battery came from), adjusted for by the back-door formula (3)
`hover`	A fictionalized HoVer claim split into sub-claims, with reports of the sub-claims from sources of varying reliability	Truth of the bridge sub-claim; a source's kind

Table 1: Item families. `source`, `diagnostic` and `hover` come in a *full* version, which states every probability, and a *qualitative* version, which gives only verbal descriptions ("usually", "most sources are careful"). Appendix A gives examples.

Log-odds treat a move from 1% to 2% as seriously as one from 33% to 50%. Near 50%, a residual of 0.1 is about 2.5 percentage points. A positive r means the direct answer is higher than the model's decomposition implies.

For causal items the target is $P(y \mid do(x))$: the probability of outcome y if treatment x were *imposed*, rather than merely observed. When Z is the only confounder and is not affected by X, the back-door adjustment gives

$$P(y \mid do(x)) \;=\; \sum_{z} P(z)\, P(y \mid x, z), \tag{3}$$

and the residual compares the model's interventional answer with its own adjustment. Normalization ($\sum_k \hat{P}(z_k) = 1$ and $\hat{P}(X) + \hat{P}(\neg X) = 1$) is checked separately. Rescaling a model's $\hat{P}(z_k)$ to sum to 1 before forming D lowers its mean residual by 0.004–0.07 log-odds (at most about a quarter, for Opus) and leaves the model ranking essentially unchanged, so the residuals reflect marginalization rather than unnormalized latent probabilities.

3.2 Item families

Table 1 summarizes the families. In the *full* versions the item is a well-posed calculation. In the *qualitative* versions the model must form its own numbers. Coherence compares a model with itself, so it is meaningful at both levels. Every item has exact ground truth: computed from the generating parameters by exact enumeration for `source` and `hover` (validated against rejection sampling) and naive Bayes for `diagnostic`, and given by the empirical frequencies for `causal`, where the table is the population. Items are filtered so that targets and latent probabilities lie in $[0.1,\ 0.9]$ and the conditionals differ across latent states by at least 0.10–0.15. Without the filter, coherence would hold trivially near 0 or 1.

The released data contain 1,500 synthetic items (300 in each of the five family–level groups: `source` and `diagnostic` at both levels, `causal` at the full level only) and 1,698 HoVer items built from 849 claims. For HoVer, Gemini 3.8 Flash decomposed each claim into logically independent sub-claims and replaced every named entity with an invented name. A second Gemini pass audited the decomposition, and Claude re-verified each accepted claim (Appendix B). The target claim is true exactly when all its sub-claims are, and sub-claim truths and source reliabilities are sampled from stated parameters, so HoVer items also have exact ground truth.

3.3 Metrics: rewording noise vs persistent incoherence

A nonzero residual has two possible sources. The model's answers may jitter with wording, making r scatter around zero, or its judgments may not fit together, making r stay put however the question is asked. The first is instability; the second is incoherence. To separate them, a subset of *noise-floor* items is asked in three wordings ("What is the probability that…", "How likely is it that…", "Estimate the probability that…").

Model each repeated residual as $r_{ik} = \mu_i + e_{ik}$ for instance i (an item × identity) and wording $k \in \{1, 2, 3\}$, where μ_i is the instance's true residual and e_{ik} is wording noise with variance σ^2. Then

$$\widehat{\sigma^2} = \tfrac{1}{n}\textstyle\sum_i \mathrm{var}_k(r_{ik}) \qquad \text{(rewording noise} = \widehat{\sigma}\text{)}, \tag{4}$$

$$\widehat{\langle\mu^2\rangle} = \tfrac{1}{n}\textstyle\sum_i \bar{r}_i^2 - \widehat{\sigma^2}/3 \qquad \text{(persistent residual} = \sqrt{\max(0, \widehat{\langle\mu^2\rangle})}\text{)}, \tag{5}$$

where $\bar{r}_i$ is the mean over wordings. The correction $\widehat{\sigma^2}/3$ removes the noise that remains in a three-wording average. The persistent residual is the root-mean-square size of the instances' true residuals. Writing $\langle\mu^2\rangle = \frac{1}{n}\sum_i \mu_i^2$ for the mean of their squares and $\bar{\mu} = \frac{1}{n}\sum_i \mu_i$ for their mean, the square of the persistent residual satisfies $\langle\mu^2\rangle = \bar{\mu}^2 + \mathrm{var}(\mu)$, with $\mathrm{var}(\mu) = \frac{1}{n}\sum_i(\mu_i - \bar{\mu})^2$, so it counts both a lean shared by all items (the **direction**, $\bar{\mu}$) and item-to-item variation (the **spread**, estimated as $\sqrt{\max(0, \mathrm{var}_i(\bar{r}_i) - \widehat{\sigma^2}/3)}$). A model whose incoherence is a fixed offset has spread near 0; one whose errors point different ways on different items has direction near 0. A noisy but coherent model scores about 0. 95% intervals come from 2,000 bootstrap resamples of instances [2].

We also report the **mean residual**, the average $|r|$ over all items using the first wording only, which reflects noise and persistent incoherence together, and the **error vs truth**, $|\mathrm{logit}\,\hat{P}(X) - \mathrm{logit}\,P(X)|$. Models are compared item by item with a paired sign-flip permutation test on mean $|r|$ (20,000 permutations), Bonferroni-corrected over all 21 model pairs.

4 Experimental setup

Sample. We sampled 20 items from each of the seven (family, level) groups and asked 60 of them in all three wordings: 3 from each full-spec synthetic group, and 12–13 from each qualitative and HoVer group, where wording effects are larger. Full-spec synthetic items are solved essentially exactly by every model, so all models except Haiku (which answered everything) were capped at 5 items in each of those three groups. All comparisons use the 95 items answered by every model: 90 with a latent total-probability identity (HoVer items carry two, over the bridge sub-claim and over a source's kind) and 5 causal items with two back-door identities each. Every model answered the same 1,590 prompts; Haiku, being uncapped, also answered 300 more full-spec prompts, which are not shared and so are left out of all comparisons.

Models and settings. Table 2 lists the models. Each prompt was an independent single-turn call with the system prompt "Answer the question using only the information provided. Give your final answer as a single probability between 0 and 1." The nominal reasoning effort was *medium*. GLM 5.3 and DeepSeek V4-Pro offer no medium setting and ran at *high*. Grok 4.7 ran at *low* because medium was projected to cost about $94; even so, it reasoned more than every model except DeepSeek. We verified that each provider honors its effort setting by comparing reasoning-token counts across levels on the same prompts. Claude, GPT and Gemini ran through their providers' batch APIs; the others ran synchronously. Responses cut off by the token limit were re-asked with a higher limit (32K for Gemini, 65K for GLM and DeepSeek); none remained truncated. A parser extracts one probability per response, handling both answer-first and answer-last layouts, and was audited by hand on every answer whose reading changed (Appendix C).

Model	Effort	Mean residual	Persistent	Noise	Spread	Dir.	Err. vs truth	Tok./ans.	Cost
GPT-6.1 Sol	med.	0.08 [0.06, 0.11]	**0.08** [0.02, 0.13]	0.16	0.08	−0.01	0.34	400	$4
Claude Opus 5.5	med.	0.12 [0.07, 0.17]	0.00 [0.00, 0.14]	0.34	0.00	+0.01	0.38	500	$18
DeepSeek V4-Pro	high	0.15 [0.10, 0.21]	**0.27** [0.10, 0.40]	0.30	0.26	+0.08	0.37	16,900	$53
Grok 4.7	low	0.21 [0.15, 0.27]	0.21 [0.00, 0.32]	0.51	0.20	+0.05	0.47	7,200	$70
Gemini 3.8 Flash	med.	0.26 [0.19, 0.34]	**0.28** [0.10, 0.40]	0.50	0.26	−0.11	0.73	4,400	$15
GLM 5.3	high	0.26 [0.20, 0.33]	0.11 [0.00, 0.25]	0.53	0.09	−0.07	0.41	6,700	$54
Claude Haiku 5.5	med.	0.37 [0.27, 0.49]	**0.35** [0.15, 0.48]	0.64	0.32	+0.14	0.50	1,100	$1

Table 2: Main results on the latent total-probability identity, in log-odds, with 95% bootstrap CIs. *Mean residual*: average $|r|$ over 90 items (first wording). *Persistent*, *Noise*, *Spread* and *Dir.* (signed mean residual): §3.3, on 83 noise-floor instances; Persistent$^2 \approx$ Spread$^2 +$ (Dir.)^2. Bold marks a persistent residual whose CI excludes 0. *Err. vs truth*: mean $|\operatorname{logit}\hat{P}(X) - \operatorname{logit}P(X)|$. *Tok./ans.*: mean reasoning tokens per answer. *Cost*: whole run (Haiku 1,890 prompts; DeepSeek at off-peak rates).

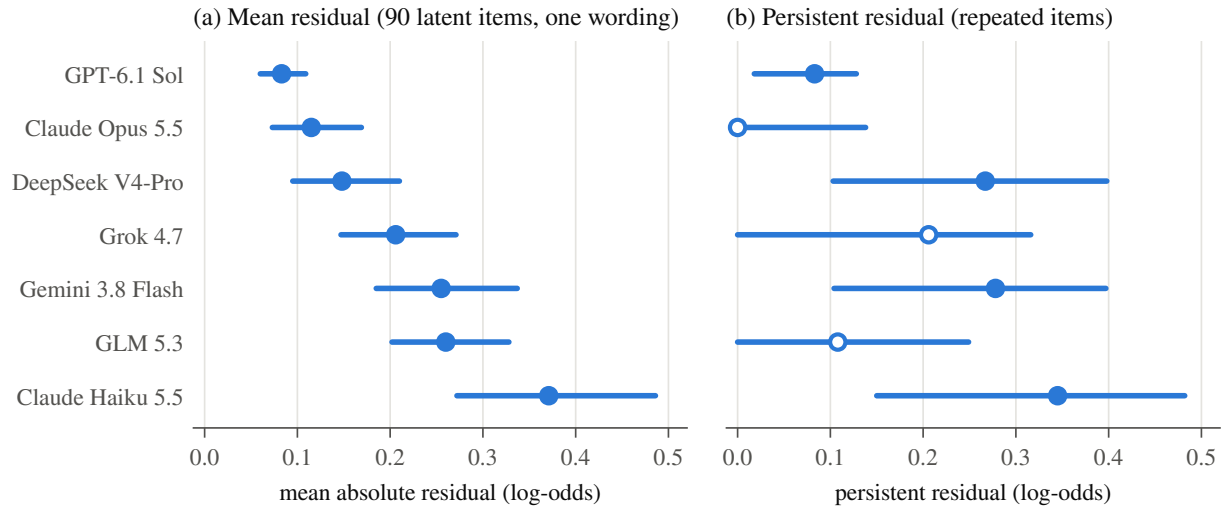


Figure 1: (a) Mean absolute residual over all 90 latent items. (b) Persistent residual on the noise-floor items. Bars are 95% bootstrap CIs; filled markers in (b) have CIs that exclude 0.

5 Results

A clear top pair, a clear bottom group, two models between. Table 2 and Figure 1 give the main results. GPT-6.1 Sol and Claude Opus 5.5 are the most coherent: their direct answers differ from their own decompositions by 0.08 and 0.12 log-odds on average, or 1.5 and 1.9 percentage points on the probability scale (medians 0.3 and 0.5). Their difference is not significant ($p_{\text{Bonf}} = 1.0$). Both are significantly more coherent than Gemini, GLM and Haiku ($p_{\text{Bonf}} \leq 0.008$), none of which can be separated from the others. DeepSeek V4-Pro (0.15) and Grok 4.7 (0.21) fall between. DeepSeek is significantly more coherent than Haiku but cannot be separated statistically from any other model. Grok is significantly less coherent than GPT ($p_{\text{Bonf}} \leq 0.001$) and more coherent than Haiku ($p_{\text{Bonf}} = 0.028$), but cannot be separated from Opus ($p_{\text{Bonf}} = 0.13$), DeepSeek, Gemini or GLM. All 21 comparisons are in Appendix D.1.

Noise vs persistent incoherence. The two residual measures can disagree sharply. GLM and Gemini have the same mean residual (0.26) and similar rewording noise (0.53 and 0.50), but GLM's persistent residual is 0.11 [0, 0.25] while Gemini's is 0.28 [0.10, 0.40]. A single answer from either is equally inconsistent, for

different reasons. GLM's residuals scatter around zero as the wording changes, so its incoherence is mostly instability that averaging several samples would reduce. Gemini's residuals vary as much with wording, but around nonzero values that persist, so its judgments themselves do not fit together, and re-asking will not fix them. For a system that relies on composing judgments, the persistent kind is the more serious failure. Four models have a persistent residual whose interval excludes 0: Haiku (0.35), Gemini (0.28), DeepSeek (0.27) and, at a small magnitude, GPT (0.08). DeepSeek is distinctive: it has the second-lowest noise (0.30), yet 44% of its residual variance is persistent, the largest share of any model. It is exact on most items (median residual 0.013) and consistently off on a few.[1]

Spread, not a shared lean. Persistent incoherence is mostly item-specific. For every model the spread nearly equals the persistent residual, and the shared lean accounts for at most a sixth of its square, except for GLM, whose small persistent residual is about 40% lean. Subtracting a per-model offset would therefore remove little: a model's judgments are off in different directions on different items. The leans that do exist are model-specific. Haiku's and DeepSeek's direct answers sit *above* their decompositions ($+0.14$ and $+0.08$ overall, rising to $+0.22$ for Haiku on full HoVer items and $+0.19$ for DeepSeek on qualitative HoVer items), and Gemini's sit *below* (-0.11 overall, -0.32 on qualitative HoVer items). On HoVer items X is a conjunction of sub-claims, so a positive lean is consistent with overestimating conjunctions relative to their parts; Gemini's opposite sign shows that this is not a universal property of language models.

Where incoherence appears. Residuals are near zero on full-spec synthetic items, which are calculations: every model's mean residual there is at most 0.055 (Table 5). They grow on qualitative items and on HoVer, but verbal evidence alone is not the main difficulty. Qualitative diagnostic items also require turning words into numbers, but have a single three-state latent, and six of seven models stay at 0.04–0.11 there (GLM 0.28). Qualitative HoVer items, where the target is a conjunction of sub-claims linked through a bridge entity, reach 0.64 for Gemini and 0.80 for Haiku, against 0.075 and 0.110 for the same two models on qualitative diagnostic items. The by-spec persistent residuals (Table 6) add model-specific detail. Gemini's is concentrated on qualitative items (0.35 [0.11, 0.51], vs 0.09 on full-spec ones), while DeepSeek's is larger on full-spec items (0.36 [0.04, 0.57]): its persistent errors arise even when every number is given. GPT's small pooled persistent residual comes almost entirely from qualitative items, where its interval excludes zero (0.11 [0.03, 0.16]). Opus's estimate there is 0.00, but its interval [0, 0.17] is too wide to separate the two. Rewording noise is higher on qualitative items for six of the seven models.

Incoherence flags inaccurate answers. Within every model, instances with larger residuals have larger errors against the truth (Table 3): Spearman correlations are 0.40–0.62, all with intervals well above zero, and the direct answer's mean error on incoherent instances ($|r| > 0.5$) is 2.2–4.8 times that on coherent ones ($|r| \leq 0.1$). Part of this is guaranteed: since $|r| \leq |\operatorname{logit}\hat{P}(X) - \operatorname{logit}P(X)| + |\operatorname{logit}D - \operatorname{logit}P(X)|$, a large residual means that the direct answer or the decomposition must be far from the truth. The residual needs no ground truth, so it is a usable lower bound on combined error. The data show which estimate breaks, and it differs by model (Table 7). For Gemini it is mostly the direct answer: on its incoherent instances the direct error is 1.68 against 1.25 for the decomposition, which is closer to the truth on 64% of them. Grok leans the same way but only slightly (1.15 vs 1.03; decomposition closer on 50%). For Haiku and DeepSeek it is the decomposition (1.41 vs 0.89; 1.09 vs 0.75), and DeepSeek's errors there sit mainly in the latent prior $\hat{P}(z)$ (1.06, vs 0.63 for the conditionals). GLM's two estimates are equally wrong (0.85, 0.86); Opus and GPT have too few incoherent instances (4 and 2) to say. Pooled over models, the conditionals are nearly exact

[1]The persistent residual is a root-mean-square over noise-floor instances, while the mean residual averages $|r|$ over all items, so a heavy tail can make the former exceed the latter, as for DeepSeek. Each measure should be compared across models rather than against the other.

	Residual vs error against truth			Residual vs reasoning tokens	
Model	Spearman ρ [CI]	Err. if $\lvert r\rvert \le 0.1$	Err. if $\lvert r\rvert > 0.5$ (n)	Across items ρ [CI]	Within item corr. [CI]
GPT-6.1 Sol	0.62 [0.45, 0.76]	0.25	0.56 (2)	−0.22 [−0.41, −0.03]	−0.05 [−0.19, 0.12]
Claude Opus 5.5	0.50 [0.33, 0.64]	0.24	0.65 (4)	−0.04 [−0.23, 0.13]	0.00 [−0.14, 0.16]
DeepSeek V4-Pro	0.53 [0.36, 0.66]	0.29	0.75 (16)	0.17 [0.00, 0.33]	−0.08 [−0.23, 0.08]
Grok 4.7	0.57 [0.40, 0.70]	0.24	1.15 (20)	0.03 [−0.15, 0.20]	0.13 [−0.07, 0.28]
Gemini 3.8 Flash	0.49 [0.30, 0.66]	0.41	1.68 (25)	−0.10 [−0.27, 0.07]	0.05 [−0.15, 0.27]
GLM 5.3	0.55 [0.39, 0.68]	0.21	0.85 (26)	0.00 [−0.18, 0.19]	−0.05 [−0.18, 0.08]
Claude Haiku 5.5	0.40 [0.23, 0.55]	0.34	0.89 (32)	0.25 [0.08, 0.41]	−0.06 [−0.23, 0.10]

Table 3: Within-model relations, over 130 latent-identity instances (first wording). *Left*: Spearman correlation of $|r|$ with the direct answer's error $|\operatorname{logit}\hat{P}(X) - \operatorname{logit}P(X)|$, and the mean error on coherent and incoherent instances. *Right*: Spearman correlation of $|r|$ with reasoning tokens summed over the identity's $1+2K$ calls; and, on the 83 noise-floor instances, the Pearson correlation of $|r|$ with log tokens after centering both within each instance, which compares the three wordings of the same question and so removes item difficulty. Its CI resamples whole instances, keeping each one's three wordings together.

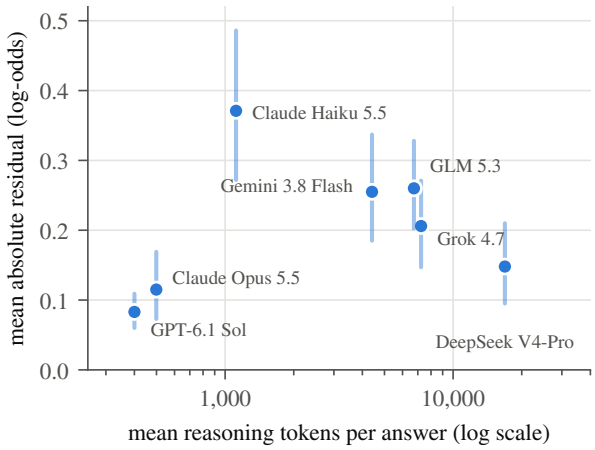


Figure 2: Mean absolute residual (95% CI) against mean reasoning tokens per answer (log scale).

on full-spec items (error 0.07) and the latent prior carries the largest error (0.13); on qualitative items every component's error is 0.63–0.72. Decomposing through the latent would thus rescue a bad holistic judgment for some models and introduce errors for others.

Reasoning length does not predict coherence. Across models (Figure 2), the two most coherent reasoned the least (about 400 and 500 tokens per answer). Grok, GLM and Gemini reasoned 9–18 times as much and were less coherent (0.21–0.26), while DeepSeek reasoned the most (about 16,900 tokens) and placed third. Seven models from six developers make this only an observation, so we also look within each model (Table 3, right). Across a model's items, the correlation between $|r|$ and reasoning tokens ranges from -0.22 (GPT, whose incoherent instances drew fewer tokens) to $+0.25$ (Haiku, whose incoherent instances drew more), and its interval excludes zero only for those two and, barely, DeepSeek. That correlation is confounded with difficulty, since a hard item can both lengthen reasoning and raise the residual. Comparing the three wordings of the same noise-floor instance holds the item fixed. There, the wording that elicited more reasoning was neither more nor less coherent: the centered correlation lies between -0.08 and $+0.13$, and its 95% interval includes zero for every model. Longer thinking on a given question neither resolves nor compounds latent incoherence.

Simple checks pass; decomposition rarely helps accuracy. Each model's probabilities over a latent's states sum to 1 (median $|\Sigma - 1| \leq 0.02$ for every model). On causal items, each model's interventional answer matches its own back-door adjustment, with mean residuals at most 0.002 except for DeepSeek (0.030) and GLM (0.047). Both exceptions are a single answer on the same item (Appendix D), and both are the classic confusion of seeing with doing. DeepSeek's $\hat{P}(y \mid do(\text{slow})) = 0.6735$ and GLM's $\hat{P}(y \mid do(\text{fast})) = 0.375$ equal the *observational* $P(y \mid \text{slow}) = 132/196 = 0.6735$ and $P(y \mid \text{fast}) = 39/104 = 0.375$ to every reported digit, while every piece of each model's own adjustment was exactly right, giving the true 0.605 and 0.275. Apart from such slips, the failures are specific to marginalizing over an inferred latent. Over all instances, a model's own decomposition D is closer to the truth than its direct answer on only 39–49% of them and improves the mean error only for Gemini, GPT and Grok, by at most 0.07.

6 Discussion and limitations

Implications. Two frontier models judge probabilities through a latent coherently enough for many purposes, within about 2 percentage points on average. Others carry inconsistencies that persist under rewording, so a system that composes their judgments, as Scientist AI does when it averages over explanations, inherits errors that sampling will not remove. Noise and persistent incoherence call for different remedies, and the persistent residual distinguishes them. Because the persistent part is mostly item-specific spread, it cannot be calibrated away with a fixed correction, and the absence of a common direction suggests these failures come from each model's training rather than a shared heuristic. The residual also has a direct practical use: it needs no ground truth, and within every model a large one marks answers with two to five times the error of coherent ones. Whether to trust the direct answer or the decomposition when they disagree, however, depends on the model.

Limitations. Several design choices limit how far the results generalize.

- **Settings.** "Medium" effort is a vendor label, not a fixed amount of reasoning, and three models deviated from it (Grok low; GLM and DeepSeek high). Temperature is not controllable for most reasoning models, and each prompt was asked once (three times on noise-floor items).
- **Sample size.** The noise-floor estimates rest on 83 latent-identity instances from 57 items, so persistent-residual intervals are wide and per-group breakdowns (Appendix D) are suggestive only.
- **Single framing.** We used one conditioning framing ("Suppose you additionally learn, from a completely reliable check, the following"); the harness's "assume" framing was not run.
- **Ground truth on qualitative items.** It comes from the hidden generating numbers. Error vs truth there partly measures how a model maps words like "usually" to numbers; coherence is unaffected.
- **HoVer construction.** Decomposition and fictionalization were done and audited by LLMs from two providers; remaining errors would affect ground truth but not coherence.
- **Scope.** We test seven API models at one point in time. Dedicated forecasting or decision models that output probabilities directly, and open-weight models run locally, are natural additions.

7 Conclusion

We introduced an eval of whether language models' probability judgments obey the law of total probability over latent variables inferred from context, separating rewording noise from persistent incoherence. Across seven frontier models, coherence varies by a factor of about 4.5. Persistent incoherence is present in several models, varies from item to item rather than acting as a fixed offset, is unrelated to reasoning length, and signals error against the truth. For proposals that build safe systems from calibrated, composable probability judgments, measuring this kind of coherence directly seems a necessary step.

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A Example items

`source`, full (item `source-full-00000`). Context, shown verbatim except that "the plant" abbreviates the plant's full name:

```
Situation: Some sources have reported on whether the Soustiakloun textile plant
will close by the end of the year.

How sources work: Every source is one of two kinds. A careful source's reports are
correct 85% of the time; a careless source's reports are correct 55% of the time.
Overall, 65% of sources are careful. A source's kind never changes, and different
reports are correct or incorrect independently of one another, given each source's kind.

Base rate: Before any reports, claims of this kind are true 30% of the time.

Track records:
| Source                  | Past reports later checked | Of those, correct |
| Trokeil Staebras        | 0                          | 0                 |
| The Vigoum Gazette      | 3                          | 1                 |
| Braedrouth Triagiabraes | 5                          | 5                 |

Current reports:
- Trokeil Staebras reports that the plant will not close by the end of the year.
- The Vigoum Gazette reports that the plant will not close by the end of the year.
- Braedrouth Triagiabraes reports that the plant will close by the end of the year.
```

Each question is appended and followed by "Respond with only a probability between 0 and 1." Queries and ground truth: $P(X) = 0.307$ ("What is the probability that the Soustiakloun textile plant will close by the end of the year?"); $P(\text{Gazette careful}) = 0.258$; $P(\text{careless}) = 0.742$; $P(X \mid \text{careful}) = 0.114$ and $P(X \mid \text{careless}) = 0.374$, asked as "Suppose you additionally learn, from a completely reliable check, the following: The Vigoum Gazette is a careful source. What is the probability that…". Check: $0.258 \times 0.114 + 0.742 \times 0.374 = 0.307$.

`hover`, qualitative (illustrative item from the gold set). Original HoVer claim: "The American publishing company headquartered in Scribner Building publishes the books of the author Donald Richard 'Don' DeLillo whose books cover subjects such as the Cold War, mathematics, politics, and economics, and global terrorism." Fictionalized sub-claims: (0) "Ostrander & Pell is an American publishing company headquartered in the Halvard Building." (1, bridge) "Ostrander & Pell publishes the books of the author Corwin Ashby." (2) "Corwin Ashby's books cover subjects such as the Cold War, mathematics, politics, economics, and global terrorism." The context describes source kinds and base rates only verbally ("A careful source's reports are almost always correct; a careless source's reports are correct only somewhat more often than a coin flip"), lists track records for three sources (among them the Baebur Bulletin), and gives five current reports on the sub-claims. The item has two latent identities: over the bridge sub-claim ($P(X) = 0.458$; $P(A_1) = 0.487$; $P(X \mid A_1) = 0.939$; $P(X \mid \neg A_1) = 0$), and over the Baebur Bulletin's kind ($P(\text{careful}) = 0.791$; $P(X \mid \text{careful}) = 0.402$; $P(X \mid \text{careless}) = 0.670$).

`diagnostic`, qualitative. "A maintenance engineer is diagnosing pump unit Paefupeirn. Exactly one of the following faults is present." Each of three causes (worn bearing, cracked impeller, clogged intake) is described verbally ("rare as a cause"; "the outlet pressure is below normal (usually)"), followed by an inspection log of checked findings. Targets: X = failure within 30 days; Y = the result of a test not yet performed. Latent: which fault is present. The data define a total-probability identity over the three-state latent for each target; the core query set used in this study asks only about X, so each item contributes one instance.

`causal` (item `causal-full-00000`). "A test lab at Siarethian recorded 500 batteries. Technicians chose the charging protocol for each battery; it was not assigned at random. Assume that the production

line is the only factor that affected both the protocol chosen and capacity retention." Counts (kept >80% capacity yes/no): older line, slow 22/86; older, fast 20/93; newer, slow 182/51; newer, fast 31/15. The interventional question reads "A battery is chosen at random from all 500 recorded batteries. Suppose that it is charged with the slow protocol, regardless of whether it came from the older production line. What is the probability that it kept more than 80% of its capacity after 500 cycles?" Truth: $P(y \mid do(\text{slow})) = 0.442 \times 0.204 + 0.558 \times 0.781 = 0.526$, while the observational $P(y \mid \text{slow}) = 0.598$ is inflated because slow charging was used mostly on newer-line batteries.

B Dataset construction

Synthetic families. `source`: base rate $\sim U(0.2,\ 0.8)$, share of careful sources $\sim U(0.3,\ 0.7)$, careful accuracy $\in \{0.85,\ 0.90,\ 0.95\}$, careless accuracy $\in \{0.50,\ 0.55,\ 0.60\}$, and per-source track-record lengths from $\{0,\ 2,\ 3,\ 4,\ 5,\ 6,\ 8,\ 10,\ 12\}$, with all reports sampled from the generative model. `diagnostic`: cause prior $\sim$ Dirichlet(3,3,3), per-cause symptom and outcome rates, and a sampled inspection log; qualitative items map rates to verbal frequency phrases. `causal`: population size in $\{200,\ \ldots,\ 600\}$, $P(z) \sim U(0.25,\ 0.75)$, confounded assignment $P(x \mid z)$, and outcome rates $P(y \mid x, z) \sim U(0.05,\ 0.95)$. All families are filtered as described in §3.2.

HoVer pipeline. From 1,500 HoVer training claims (500 each of 2, 3 and 4 hops), Gemini 3.8 Flash judged whether each claim is a pure conjunction and, if so, produced 2–5 logically independent, self-contained sub-claims with exact negations, a consistent entity map to invented names, and a bridge index. The instructions require that descriptive modifiers be asserted in exactly one sub-claim, that nothing be dropped, and that comparisons, disjunctions and counting claims be rejected; five hand-written gold decompositions serve as few-shot examples. A second Gemini pass audited each decomposition for conjunction equivalence, independence, exact negations, full fictionalization and self-containment. The 908 accepted claims were re-verified by Claude through the Message Batches API, and the 849 that passed form the released set. 238 HoVer items whose invented names look alike are flagged (`--exclude-similar-names`). Generation cost about $45 (Gemini) and $6 (Claude).

C Model settings, cost and parsing

Model	Provider	Effort (levels offered)	Mode	Prompts	Reasoning tok./ans.	Cost (USD)
Claude Haiku 5.5	Anthropic	medium	batch	1,890	1,117	0.62
Claude Opus 5.5	Anthropic	medium	batch	1,590	499	17.61
GPT-6.1 Sol	OpenAI	medium	batch	1,590	400	3.96
Gemini 3.8 Flash	Google	medium	batch	1,590	4,409	14.52
Grok 4.7	xAI	low (low/med./high/xhigh)	sync	1,590	7,236	70.44
GLM 5.3	Z.ai	high (low/high/max)	sync	1,590	6,733	53.57
DeepSeek V4-Pro (0813)	DeepSeek	high (none/high/max)	sync	1,590	16,872	≈53

Table 4: Settings and cost. DeepSeek's cost is at off-peak rates ($106.87 at peak rates). Batch runs are billed at half price.

Answer parsing. Most models answer with the number alone or explain and end with it; GLM, and Haiku at times, state the answer first in bold and explain afterwards, and explanations contain other numbers in $[0,\ 1]$. When a response contains several distinct candidates, the parser takes, in order: (1) a last line consisting

only of a probability; (2) a first line consisting only of a probability, optionally labeled ("Answer: 0.9"); (3) the number after "answer is" or "final answer"; (4) otherwise the last number. All answers are re-parsed at analysis time so that every model is scored by the same rules. Relative to the values recorded at run time, 125 Haiku answers and 4 GLM answers changed, and every one was checked by hand against its text. No answer from the other five models changed.

D Additional results

Group (n identity instances)	Haiku	Opus	DeepSeek	Gemini	GLM	GPT	Grok
diagnostic / full (5)	0.002	0.006	0.000	0.000	0.002	0.000	0.002
source / full (5)	0.020	0.045	0.001	0.013	0.043	0.000	0.055
diagnostic / qualitative (20)	0.110	0.055	0.094	0.075	0.275	0.044	0.086
source / qualitative (20)	0.382	0.176	0.078	0.261	0.336	0.152	0.224
hover / full (40)	0.376	0.046	0.186	0.169	0.109	0.012	0.234
hover / qualitative (40)	0.798	0.230	0.307	0.641	0.438	0.164	0.368
causal back-door (10)	0.001	0.002	0.030	0.000	0.047	0.000	0.000

Table 5: Mean absolute residual (log-odds) by group, first wording.

	Full spec (32 instances)		Qualitative (51 instances)	
Model	Noise	Persistent	Noise	Persistent
Claude Haiku 5.5	0.37	0.34 [0.00, 0.57]	0.77	0.35 [0.00, 0.52]
Claude Opus 5.5	0.07	0.03 [0.00, 0.06]	0.43	0.00 [0.00, 0.17]
DeepSeek V4-Pro	0.28	**0.36** [0.04, 0.57]	0.31	0.19 [0.00, 0.31]
Gemini 3.8 Flash	0.40	0.09 [0.00, 0.20]	0.56	**0.35** [0.11, 0.51]
GLM 5.3	0.45	0.00 [0.00, 0.00]	0.57	0.18 [0.00, 0.33]
GPT-6.1 Sol	0.12	0.01 [0.00, 0.01]	0.18	**0.11** [0.03, 0.16]
Grok 4.7	0.63	0.27 [0.00, 0.46]	0.41	0.15 [0.00, 0.25]

Table 6: Rewording noise and persistent residual by spec level (noise-floor items). Bold: CI excludes 0.

	Coherent ($\lvert r\rvert \leq 0.1$)					Incoherent ($\lvert r\rvert > 0.5$)					
Model	n	$\hat{P}(z)$	$\hat{P}(X \mid z)$	$\hat{P}(X)$	D	n	$\hat{P}(z)$	$\hat{P}(X \mid z)$	$\hat{P}(X)$	D	D closer
Claude Haiku 5.5	53	0.36	0.32	0.34	0.34	32	1.06	1.04	0.89	1.41	28%
Claude Opus 5.5	91	0.35	0.29	0.24	0.25	4	1.61	0.51	0.65	1.19	25%
DeepSeek V4-Pro	91	0.32	0.32	0.29	0.30	16	1.06	0.63	0.75	1.09	38%
Gemini 3.8 Flash	63	0.37	0.33	0.41	0.42	25	0.88	0.90	1.68	1.25	64%
GLM 5.3	59	0.25	0.21	0.21	0.23	26	0.76	0.55	0.85	0.86	50%
GPT-6.1 Sol	94	0.29	0.28	0.25	0.25	2	0.50	0.67	0.56	0.12	100%
Grok 4.7	73	0.26	0.23	0.24	0.25	20	0.60	0.79	1.15	1.03	50%

Table 7: Error attribution: mean log-odds error against the truth of each estimate (averaged over latent states for $\hat{P}(z)$ and $\hat{P}(X \mid z)$), for coherent and incoherent latent-identity instances (first wording). D: the model's own decomposition. *D closer*: share of incoherent instances where D is closer to the truth than $\hat{P}(X)$. Pooled over models, errors on full-spec instances are 0.13 ($\hat{P}(z)$), 0.07 ($\hat{P}(X \mid z)$), 0.09 ($\hat{P}(X)$) and 0.11 (D); on qualitative instances, 0.69, 0.63, 0.69 and 0.72.

Causal anomalies. The only back-door residuals above 0.01 come from item `causal-full-00244` (300 batteries; older line: slow 110/29, fast 3/20; newer line: slow 22/35, fast 36/45 kept/did not keep >80% capacity). DeepSeek answered 0.6735 for $P(y \mid do(\text{slow}))$ and GLM answered 0.375 for $P(y \mid do(\text{fast}))$. These are exactly the observational rates among slow- and fast-charged batteries. Both models answered $P(z)$ (0.54, 0.46) and every $P(y \mid x,z)$ correctly, so their own adjustments equal the truth (0.605 and 0.275), and the residuals (+0.30 and +0.46) are entirely the direct interventional answer ignoring the confounder. On the other arm of the same item, both models' interventional answers matched their own adjustments ($|r| \leq 0.01$).

Model	Err. direct	Err. decomposition	Decomp. closer	Normalization (median $\lvert\Sigma - 1\rvert$, all checks)
Claude Haiku 5.5	0.50	0.64	44%	0.010
Claude Opus 5.5	0.38	0.42	43%	0.003
DeepSeek V4-Pro	0.37	0.44	40%	0.000
Gemini 3.8 Flash	0.73	0.66	49%	0.010
GLM 5.3	0.41	0.44	39%	0.020
GPT-6.1 Sol	0.34	0.33	47%	0.000
Grok 4.7	0.47	0.46	47%	0.000

Table 8: Accuracy against ground truth (mean log-odds error over 130 latent-identity instances) for the direct answer and for the model's own decomposition, the share of instances where the decomposition is closer, and the median deviation from 1 pooled over all normalization checks.

D.1 Pairwise comparisons

A	B	Δ	p_{Bonf}	A	B	Δ	p_{Bonf}
Haiku	Opus	0.256	≤0.001	DeepSeek	Gemini	−0.108	0.303
Haiku	DeepSeek	0.224	≤0.001	DeepSeek	GLM	−0.112	0.168
Haiku	Gemini	0.116	0.541	DeepSeek	GPT	0.065	0.639
Haiku	GLM	0.112	0.819	DeepSeek	Grok	−0.058	1.000
Haiku	GPT	0.289	≤0.001	Gemini	GLM	−0.004	1.000
Haiku	Grok	0.166	0.028	Gemini	GPT	0.173	≤0.001
Opus	DeepSeek	−0.032	1.000	Gemini	Grok	0.050	1.000
Opus	Gemini	−0.140	0.008	GLM	GPT	0.177	≤0.001
Opus	GLM	−0.144	0.002	GLM	Grok	0.054	1.000
Opus	GPT	0.033	1.000	GPT	Grok	−0.123	≤0.001
Opus	Grok	−0.090	0.132				

Table 9: Paired item-level differences Δ = mean $|r|_A$ − mean $|r|_B$ over 90 latent items, with sign-flip permutation p-values (20,000 permutations) Bonferroni-corrected over 21 pairs. "≤0.001" is the resolution floor of the test after correction.

D.2 Robustness to clipping, scale and normalization

	Answers (%)		Mean $\lvert r\rvert$, clip at			$\lvert\hat{P}(X)-D\rvert$ (pp)		Latent states	
Model	clipped	exactly 0/1	0.001	0.01	0.05	mean	median	renorm. $\lvert r\rvert$	med. $\lvert\sum_k \hat{P}(z_k)-1\rvert$
GPT-6.1 Sol	7.4	5.7	0.084	0.083	0.083	1.5	0.3	0.071	0.000
Claude Opus 5.5	7.9	2.5	0.123	0.115	0.110	1.9	0.5	0.085	0.003
DeepSeek V4-Pro	8.2	5.8	0.162	0.148	0.125	2.1	0.3	0.143	0.000
Grok 4.7	7.7	6.1	0.209	0.206	0.192	3.0	1.4	0.188	0.001
Gemini 3.8 Flash	9.6	7.1	0.371	0.255	0.206	3.5	1.6	0.251	0.012
GLM 5.3	6.7	5.8	0.265	0.260	0.252	4.9	2.4	0.224	0.020
Claude Haiku 5.5	6.6	5.3	0.378	0.371	0.348	6.1	3.3	0.302	0.013

Table 10: Robustness. *Clipped*: share of all 1,590 answers at or beyond 0.01 or 0.99. *Mean* $|r|$: the headline measure (item-level mean, latent identities) with clip bounds $[\varepsilon,\ 1-\varepsilon]$. $|\hat{P}(X)-D|$: the residual on the probability scale, in percentage points. *Latent states*: mean $|r|$ after rescaling the $\hat{P}(z_k)$ to sum to 1 (item-level mean over 90 items, like the headline measure), and the median deviation of their sum from 1 over the 130 latent-identity instances (first wording). The latter's base differs from the normalization column of Table 8, which pools all of the harness's normalization checks, so the medians differ slightly.

The model ranking is the same at clip levels 0.01 and 0.05 and on the probability scale. At 0.001, Gemini, which gives the most answers of exactly 0 or 1, rises from 0.26 to 0.37 and swaps places with GLM; the two are within 0.005 at the default clip. Renormalizing $\hat{P}(z)$ lowers mean residuals by 0.004–0.07 (most for Haiku, 0.37 to 0.30; proportionally most for Opus, 0.115 to 0.085) and again only swaps Gemini and GLM, so the residuals reflect marginalization rather than latent probabilities that fail to sum to 1.

E Estimator details

For instance i with wording residuals r_{i1}, r_{i2}, r_{i3}, write $r_{ik} = \mu_i + e_{ik}$ with $\mathbb{E}[e_{ik}] = 0$ and $\mathrm{var}(e_{ik}) = \sigma^2$. The sample variance across wordings is unbiased for σ^2, so its average over instances estimates σ^2. The wording mean $\bar{r}_i = \mu_i + \bar{e}_i$ has $\mathbb{E}[\bar{r}_i^2] = \mu_i^2 + \sigma^2/3$, so $\frac{1}{n}\sum_i \bar{r}_i^2 - \widehat{\sigma^2}/3$ is unbiased for $\langle\mu^2\rangle = \frac{1}{n}\sum_i \mu_i^2$. Since $\langle\mu^2\rangle = \bar{\mu}^2 + \frac{1}{n}\sum_i(\mu_i - \bar{\mu})^2$, the persistent residual combines the shared direction $\bar{\mu}$ with item-to-item variation; the harness also reports the spread-only quantity (`systematic_spread_sd`), which excludes $\bar{\mu}$. The square root of a truncated unbiased estimate is biased upward when the true value is near zero, which is why an interval that excludes 0, not the point estimate, is the relevant evidence of persistent incoherence.